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Ethical Considerations for Human Biology Research

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The past 25 years have seen an ever increasing emphasis on and discussion of ethics in professional life. The Center for the Study of Ethics in the Professions at the Illinois Institute of Technology currently has a library of over 850 Codes of Ethics for various professions. Professional societies often have ethics modules online. Courses on ethics or ethics training are recommended parts of graduate curricula. Medicine, law, engineering, and business all have ethical standards and codes. The scientific community as a whole also shares a set of guiding principles that have been codified into a code of ethics for research and practice. In addition, each academic discipline has its own set of standards and principles, since each discipline has its own history and its own ethical dilemmas. Here I will briefly review ethical principles common to the scientific community as well as some of the ethical dilemmas faced by human biologists. This is not a comprehensive account. I direct the reader to the volume *Biological Anthropology and Ethics: from Repatriation to Genetic Identity* (Turner, 2005a) for a fuller discussion of the issues presented here.

Codes of ethics exist because every individual faces choices. These codes provide a framework for making informed choices in situations where there are conflicting obligations and responsibilities. The codes provide a framework of general principles for discussion and choice. No code can anticipate each unique situation. Discussion and reflection are vital to anticipate situations that may require quick decisions. Anthropologists (as evidenced in the American Anthropological Association [AAA] Code of Ethics, the American Association of Physical Anthropologists [AAPA] Code of Ethics) recognize a series of responsibilities – to the people with whom they work and whose lives they study, to scholarship, to science, to the public, to students and trainees, to employers, and employees. With these multiple levels of responsibility it can be difficult to determine which takes precedence in a given situation. Linda Wolfe (2005) has reviewed the responsibilities that anthropologists face that are common to all scientists in the practice of their science. These

include: avoid falsifying data or plagiarizing; avoid carelessness when collecting data; avoid falsifying grant records; avoid mistreating or discriminating against others, specifically students, coworkers, and employees; avoid giving professional advice on topics you are not qualified to discuss; avoid falsely representing a professional organization; report conflicts of interest; avoid clandestine research that cannot be published; follow rules of multiple authorship and be an objective peer reviewer. These are well established and agreed upon. However, the most difficult responsibilities a physical anthropologist or human biologist faces are often to the people we study. Discussion of these responsibilities can be subsumed under a general discussion of bioethics.

Bioethics, a special branch of applied ethics, is concerned with human health and human subjects research. Bioethics sets forth standards and principles that have become the model for work in medicine and research. Formal bioethics began after World War II, in the wake of Nazi experimentation, with the Nuremberg Code. This Code sets forth explicitly the principle of voluntary consent and lists criteria that must be met before any experimentation can be done on human subjects (Turner, 2005b). In the decades following Nuremberg, several ethical codes were enacted by the US government, the National Institutes of Health, the World Medical Association, and the Department of Health, Education and Welfare. In 1974, Congress enacted the National Research Act, which mandated an Institutional Review Board (IRB) review for all Public Health Service-funded research, and authorized the establishment of the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. The Commission produced a document, known as the Belmont Report. The Belmont Report articulated three ethical principles: autonomy or respect for persons, beneficence, and justice. These principles are usually understood as do no harm, apply the rules of justice and fair distribution, do not deprive persons of freedom and help others (for a fuller discussion, see Stinson, 2005). The

Belmont Report has been codified into federal regulations and is used by Institutional Review Boards (IRBs) in their analysis of research protocols. These IRBs are local and found at institutions conducting or supporting human subjects' research. Institutional Review Boards are responsible for the review and approval of research activities involving human subjects. Their primary mandate is to protect the rights and safeguard the welfare of human research subjects. In 1981, final Department of Health, Education, and Welfare (DHEW) approval was given in 45 CFR 46, Subparts A, B and C (Title 45 Public Welfare, Code of Federal Regulations, Part 46 Protection of Human Subjects, 1991). On March 18, 1983, Subpart D was added to the regulations, providing additional protections for children who are subjects in research. Initially the Department of Health and Human Services (DHHS, the agency that replaced DHEW) regulations applied only to research conducted or supported by DHHS. But, in June 1991, the United States published a common policy for federal agencies conducting or supporting research with human subjects. That policy, which is known as "the Common Rule," extended the provisions of 45 CFR Part 46, to 14 other federal agencies; it now governs most federally supported research. The composition and operation of each university or institution IRB must conform to the terms and conditions of 45 CFR Part 46. (NIH Human Research Protection Program, <http://www1.od.nih.gov/oma/manualchapters/intramural/3014/>). Since the establishment of the IRB system, other federal commissions, including the National Research Council, the National Bioethics Advisory Commission, and the President's Council on Bioethics, have continued to examine issues concerning human subjects and to prepare updated guidelines. Human subjects research must be overseen by local IRBs. Funding by federal agencies will not be approved without IRB oversight and approval. In multi-institution or multi-national projects more than one IRB may be involved. Since every institution in this country has its own IRB and every country may have its own regulations, approval to do research can be cumbersome. But as Long (2005, p. 278) states, "As a general rule, investigators should simultaneously meet the highest standards of both our own culture and those of the research subjects' culture." According to Long (2005, p. 279) the current guidelines now "mandate that among other things, the researcher is responsible for proper scientific design, monitoring participant rights and welfare in the course of research and ensuring that all personnel on the research team are qualified and trained in human subjects protections."

Until 45 CFR 46 was implemented, oversight of research projects was not well codified. It is certainly possible to look back at research conducted in the years between the Nuremberg Code and 45 CFR 46 and

identify practices that would no longer be considered wholly acceptable. However, these practices may have been fully acceptable and even far-sighted at the time research was conducted. Recently a controversy occurred surrounding James Neel and his research among the Yanomami. The controversy erupted a short time before a book by Patrick Tierney, *Darkness in El Dorado* (2000), was published. In proofs of the book, Tierney had accused Neel of starting a measles epidemic by injecting local villagers with a virulent measles vaccine. These charges were withdrawn before the book was published, due to a huge outcry by the scientific community about the validity of these claims. But controversy continued, with some researchers claiming Neel and his team did not do all they could to alleviate the measles epidemic among the Yanomami. Several professional organizations, including the AAA, set up task forces to review all materials. Within the anthropological community the controversy quickly came to concern the seeming conflict between obligations to science and humanitarian efforts. Those members of the task force charged with reviewing the Neel material (Turner and Nelson, 2005) found that Neel worked very hard to alleviate the measles epidemic he found in Venezuela.

There were other issues in the *Darkness in El Dorado* controversy that continue to have resonance for researchers today. These include the nature of informed consent, reciprocity for samples, and disposition of samples.

INFORMED CONSENT

There are several excellent reviews of the history of informed consent. Philosophers, ethicists, historians of science, and attorneys have all written about this (see for example, Beauchamp and Childress, 1989; Gert et al., 1997). Before the Nuremberg Report, most clinical medicine researchers were guided by the principle of beneficence and dealt little with the principle of autonomy. This principle of autonomy or respect for persons, articulated as voluntary or informed consent, was of primary importance in the shaping of the Nuremberg Code, which resulted as a response to Nazi medical experimentation. The Nuremberg Code and others that followed presented an ideal for dealing with human subjects. However, the particulars of application of this ideal to real-life situations was not as well articulated. There were certainly also situations and research projects conducted before the current regulations when principles of informed consent were missing (one of the most egregious examples being the Tuskegee Syphilis Study, which ended in 1972). The real question in many studies conducted before the implementation of the Belmont Report is how informed was informed consent. How well articulated were the goals, methods, and

consequences of the research? While these questions are important when dealing with relatively informed Western, English-speaking individuals, how were they handled with non-Western, non-English-speaking, indigenous populations?

Recognizing this as a special case, the World Health Organization (WHO) convened a working group which met in 1962 and 1968 to discuss studies of “long-standing, but now rapidly changing, human indigenous populations” (Neel, 1964). Two reports were produced, both authored by James Neel (1964, 1968), which detailed the relationship and ethical obligations of researcher to study population. Neel particularly emphasized six factors of special importance: (1) The privacy and dignity of an individual must be respected and anonymity of subjects must be maintained. (2) Satisfactory, but carefully considered, recompense should be given for participation in a study. (3) The local population should benefit from the study by medical, dental, and related services. (4) Attempts should be made to maintain congenial social relationships with participants. (5) Learned individuals should be consulted. (6) There should be the utmost regard for cultural integrity of the group.

It is clear these principles were in place during the heyday of studies conducted under the Human Adaptability Section of the International Biological Programme (Collins and Weiner, 1977). Informed consent was sought, but not in the ways it is now sought. Turner and Nelson (2002, 2005) conducted a survey of 14 researchers working in the field during this time. They asked specifically how information was conveyed to individuals involved in studies. Every survey respondent stressed that there were individuals in the populations they worked with who did not participate. Voluntary consent was therefore assumed. Researchers either had government or local permission to conduct their studies. In every case, researchers gave some explanation of the motivation for the study. But some of these explanations were not necessarily complete. Researchers felt that perhaps local populations might not understand precisely the questions they were pursuing. Scientists who were part of the Yanomami expedition in the late 1960s have stated that the Yanomami were told that the researchers were going to look for diseases of the blood. This was true, but there were other things that were researched as well. Some Yanomami that have spoken to outsiders after the publication of the Tierney book have stated that there was an expectation of greater medical benefit from the work.

GROUP CONSENT

Physical anthropologists and human biologists are in a unique position – they are interested in the range of human variation and they frequently study traits that

require them to return again and again to local, identified communities. Some researchers have had multi-decade relationships with their study populations. Over the decades, standards of what is included in informed consent have changed. Friedlaender (2005) gives a detailed account of his 35-year relationship with groups in the Solomon Islands and the changing standards of informed consent that he has implemented in his work. The current standard is, of course, full disclosure of the research project and the risks and benefits. This includes returning to the population for additional consent if samples might be used for a related, but not identical project. The best way to describe the current paradigm is in terms of an ongoing relationship between subjects and researchers, with subjects as active participants in research design and implementation.

Researchers are conditioned to think about the impact of research on an individual – on his or her health or psychological well being. It is important now that the researcher think about the impact of the research on the study population. The Belmont Principles protect individual participants in research projects. But many anthropological studies are population based and the findings of these studies can impact and affect whole populations. Consultation and group consent is now sought from populations. But group consent leads to a new suite of questions (enumerated by Juengst, 1999; Turner, 2005b): “Who speaks for the group? If the group is nested within a larger group, who represents the original group? What are the limits of the group? What is the relationship between expatriate groups and the community of origin? Does permission from a national government to conduct research have meaning for the community being studied? How does one obtain informed consent from an individual or a group whose members have little understanding of the project or the risks involved? How can the culture of the population be taken into account in the design or implementation of the project? What are the implications concerning the disclosure of the identity of the group? Can consent be withdrawn sometime in the future? How? Can samples be withdrawn sometime in the future? How? Are there appropriate benefits for the population under study?” This series of questions must be asked by every researcher engaged in research with human populations. In fact, these same questions are asked by cultural anthropologists, archaeologists, skeletal biologists, and any other researcher working with identified human populations.

An example of group consent and consultation can be found in the work of O’Rourke et al. (2005) who have been engaged in ancient DNA research with several populations. Each of the populations O’Rourke has worked with necessitated an individualized approach for access to samples. Some communities

requested in-person meetings; others did not. Different communities had different restrictions on the size of samples. Working with Paleo-Indian remains for ancient DNA (aDNA) or skeletal biology studies in this country requires adherence to Native American Graves and Repatriation Act (NAGPRA) regulations. This may mean that some of these studies cannot take place. On the other hand, some scientists (Larsen and Walker, 2005) have been able to open discussions with native peoples on the study and disposition of human remains.

WEIGHING RISKS AND BENEFITS

Distinctions are made in the Belmont Report between biological and behavioral research. In biological or medical research, risks can often be more clearly identified than in behavioral research. But, behavioral research can cause emotional, psychological, or social harm (Stinson, 2005). Embarrassment or social stigma can be real consequences of participation in a research project. An individual may find questions embarrassing or might face social consequences if his or her answers to questions were known. One of the most important risks to an individual is disclosure of identity. Institutional Review Boards are very aware of the risks of this disclosure and will look closely at the ways in which identity can be safeguarded.

Winston and Kittles (2005) describe the challenges to perceived identity that were sometimes generated by the African Ancestry Project. Williams (2005) also discusses some disclosure issues faced by descendants of Thomas Jefferson after a study of DNA from descendants of Sally Hemings and the Jefferson family. Stigmatization can also occur at the group level and this may be especially true for marginalized or identified populations. Members of a group might be stigmatized by having their circumstances discussed. How can one avoid this situation? Researchers feel that a frank and full discussion of this risk can lead to a negotiation between subject and researcher on the presentation of the results and the naming of the group as participant. Williams (2005) also suggests that constant vigilance during the planning and execution of the project be paramount.

COMPENSATION

There is often a huge differential between the researcher and the participant in studies in education, socioeconomic status and access to resources. Researchers can and do compensate participants in research studies for their time and effort. But the compensation must not be so great as to compel participation in a study. In addition, this differential may also influence rapport

and trust between the investigator and the participant. In medical studies in this country, compensation may take the form of some level of medical care. However, what are appropriate compensations for research studies conducted with non-Western, identified populations? If a study includes medical personnel, some level of medical care may be given to the participants. But this is not necessarily the type of care individuals need. Certainly there are some conditions where antibiotics or analgesics can be useful and even life saving. What if a person is identified as diabetic? A single visit from a medical professional will not be sufficient to help this person. Referrals to more long-term care facilities may be in order. In the past, researchers have given many items as compensation. Researchers usually select these items in consultation with those familiar with the culture. Food items, photos, tools, machetes, and cash have all been given as compensation. Other items have been given to the group or community. One of the more recent examples of compensation involved technology transfer and training of individuals to use this technology (Bamshad, 1999; Jorde, 1999).

DATA SHARING

Circular A-110 of the US Office of Management and Budget stipulates that data collected through grants awarded by federal agencies such as National Institutes of Health (NIH) and National Science Foundation (NSF) are public. Federal agencies encourage the broad and rapid dissemination of information throughout the scientific community reflecting the scientific ideal of an open community of scholars pursuing novel ideas and avenues of research. Major complex electronic databanks already exist. Genetic information is shared via the International Nucleotide Sequence Database, which includes GenBank, the DNA DataBank of Japan, and the European Molecular Biology Laboratory. The NSF has a database initiative. Data used by physical anthropologists, however, is often unique and difficult to obtain. It may not be possible to obtain second sets of blood or saliva samples or measurements, or interviews from members of identified communities. Individual and group consent and confidentiality become major issues if samples are shared. Specific questions about data sharing range from the definition of data to fair use for the individual collecting the data (Turner, 2005c). The physical anthropology program of the NSF has a data-sharing requirement. However, the design implementation of this requirement is up to the individual researcher, but must go beyond publication of results in a scientific journal. Questions related to the ethics and the requirements of data sharing are really just beginning in our community.

IMPLEMENTING ETHICAL STANDARDS

Discussions of ethics usually generate more questions than answers. Field-work situations are unique. Issues change from place to place and from time to time. Ethical standards have also changed over time. There are now laws and standards concerning research and research subjects. In the vast majority of cases, researchers make their best possible efforts to behave according to the letter and the spirit of the law. But there are judgment calls and the IRB system is in place for oversight – so that someone else can add perspective. It is important, though, that ethics not be confined merely to IRB oversight. Continual discussion, especially during training, is important. Many training programs offer specialized courses in ethics. Others include discussion of ethics in every class. In either case, students and faculty need to continually engage each other in an assessment of ethical standards in professional life.

DISCUSSION POINTS

1. In the digital age, the treatment, storage, and transportation of sensitive data are particularly important. Storage of sensitive personal information on a laptop may not be secured. What precautions must be taken to secure this type of information?
2. How can one overcome the logistical issues associated with obtaining additional informed consent from remote populations?
3. What if a previously unknown condition is revealed during genetic screening? In many areas of the world, referring a subject to a physician is not feasible. What do you do?
4. The disparity in wealth between a researcher and a participant may influence rapport between the two. What are some methods to deal with this?
5. There may be situations when a research agenda and humanitarian concerns come into conflict. Discuss the expectations and limits of a researcher's responsibility to the participant community.

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